

Gourishankar Mahadeo Bansode

Tempe, AZ

+1-602-491-8770

gourishankar@asu.edu

[linkedin.com/in/gourishankarb/](https://www.linkedin.com/in/gourishankarb/)

github.com/gourishankar27

gshankar.me

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Hiring Manager

Waymo

Mountain View, CA

Dear Hiring Manager,

I am writing to express my interest in the **Senior Software Engineer, Planner Evaluation** role at Waymo. With over two years of experience building backend and data-intensive systems in Python and C/C++, and as a current M.S. student in Robotics and Autonomous Systems (Artificial Intelligence) at Arizona State University, I am excited about the opportunity to help measure and improve the driving behavior of the Waymo Driver. Working on quantitative signals, simulation, and evaluation for a safety-critical autonomous system is exactly where I want to focus my career.

At Tata Advanced Systems Limited, I worked on large, distributed defense platforms with strict reliability and performance requirements. I led backend development for the Indian Army's Integrated Electronic Warfare Systems (IEWS), building data pipelines and services for communication and configuration modules across heterogeneous hardware while coordinating a team of six engineers. I also designed and documented a reusable Health Monitor SDK for Indian Navy radar and combat management projects (3D-ASR, CMS), exposing standardized health and performance signals and fault detection interfaces across multiple products. This work required navigating and improving a large shared C/C++ and Python codebase, enforcing code health, and treating metrics, logs, and observability as first-class tools for understanding complex system behavior.

A recurring theme in my experience has been using data and metrics to close the loop between software changes and system behavior. On defense fire-control projects such as Dhanush (Howitzer), ATAGS (Howitzer), and Pinaka (Multi-Barrel Rocket Launcher), I helped automate mission-critical firing workflows, reducing manual intervention by 75%. I instrumented the software with signals for timing, error modes, and operator actions, and analyzed logs to detect regressions and improve reliability through deterministic control and repeatable test procedures. Similarly, in my role as a Software Engineering Intern at Softnautics, I developed a data-driven, gesture-implied car infotainment prototype using OpenCV and TensorFlow, integrating REST APIs for real-time data exchange and achieving 92% gesture recognition accuracy. I designed evaluation datasets, iterated on the model based on quantitative results, and used profiling and debugging to resolve performance bottlenecks in the pipeline.

My graduate work further aligns with the Planner Evaluation team's focus on rich data, simulation, and quantitative evaluation. Most recently, in a project on *Vision-Guided Differentiable Physics for Robotic Manipulation*, I developed a unified perception-to-control framework that integrates

3D Gaussian Splatting (3DGS) with NVIDIA Warp for real-time differentiable physics-based manipulation. I engineered a synthetic data pipeline in Isaac Lab to generate high-fidelity RGB-D datasets, automatically labeling object dynamics and contact forces. On top of this, I implemented custom differentiable physics kernels for system identification, inferring latent physical properties such as friction and mass directly from visual gradients and optimizing these properties to reduce discrepancies between simulated and observed behavior. This work has given me hands-on experience defining and validating metrics over simulated trajectories, building tight feedback loops between perception, physics-based models, and downstream control policies.

In parallel, I have worked on projects that emphasize time-series modeling and autonomous control. For example, I developed a time-series forecasting framework using Transformer architectures (Temporal Fusion Transformer, LSTM, and ARIMAX) to predict monthly S&P 500 index movements, integrating macroeconomic indicators and standardizing evaluation metrics and visualizations to compare models fairly on held-out test sets. I also built an autonomous drone landing system in MATLAB/Simulink, combining computer vision with feedback control and validating performance using metrics such as detection reliability, tracking error, and landing tolerance in both simulation and hardware.

These experiences map well to the responsibilities of the Planner Evaluation team: developing meaningful signals to characterize safety, comfort, and efficiency; using simulation and logs to measure the impact of planner and stack changes; and improving the connection between onboard software modifications and simulated outcomes. I enjoy diving into complex systems, building data pipelines and evaluation tooling, and collaborating with engineers, data scientists, and researchers to turn ambiguous questions into clear, quantitative analyses.

I am confident that my background in safety-critical systems, data and metrics, differentiable simulation, and robotics and machine learning would allow me to contribute quickly to the Planner Evaluation team. I would be excited to bring my experience with Python, C/C++, data processing pipelines, and quantitative evaluation to help Waymo scale a safe, trusted, and highly measurable autonomous driving system.

Thank you for considering my application. I would welcome the opportunity to discuss how I can contribute to Waymo's mission.

Sincerely,

Gourishankar Mahadeo Bansode